

A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps

Xiaowen Jiang¹ Valerio Cambarelli^{2*} Gianluca Agresti³
Cynthia Ifeyinwa Ugwu⁴ Adriano Simonetto⁴ Fabien Cardinaux³ Pietro Zanuttigh⁴

¹ EPFL, Switzerland ² Sony Depthsensing Solutions NV, Belgium

³ Sony Europe B.V., R&D Center, Stuttgart Laboratory 1, Germany ⁴ University of Padova, Italy

Abstract

Sparse active illumination enables precise time-of-flight depth sensing as it maximizes signal-to-noise ratio for low power budgets. However, depth completion is required to produce dense depth maps for 3D perception. We address this task with realistic illumination and sensor resolution constraints by simulating ToF datasets for indoor 3D perception with challenging sparsity levels. We propose a quantized convolutional encoder-decoder network for this task. Our model achieves optimal depth map quality by means of input pre-processing and carefully tuned training with a geometry-preserving loss function. We also achieve low memory footprint for weights and activations by means of mixed precision quantization-at-training techniques. The resulting quantized models are comparable to the state of the art in terms of quality, but they require very low GPU times and achieve up to 14-fold memory size reduction for the weights w.r.t. their floating point counterpart with minimal impact on quality metrics.

1. Introduction

Time-of-flight (ToF) sensors are active depth sensing devices [53] with the potential to provide more reliable scene understanding by true 3D perception. Due to their low power consumption and accuracy at real-time frame rates, ToF sensors were recently integrated in mobile consumer devices [33]. However, ToF relies on active illumination [53] which accounts for a relevant part of its power consumption. To use the limited power budget of a mobile device more efficiently, the scene can be illuminated with a dot pattern light source so that its radiant intensity concentrates onto a small number of regions (*dots*). A low-power ToF sensor for indoors 3D perception typically captures 500 ~ 1500 dots per frame. Because of this sparsity level, sensor fusion techniques are necessary to obtain dense

depth maps. Hereafter, we tackle this *depth completion* task by using a single RGB image frame together with a sparse depth map warped to the RGB camera view (at small baseline of a few *mm* to minimize occlusion issues w.r.t. the ToF sensor). We dub this the *sparse ToF* setting.

To obtain dense depth from a sparse depth map, deep learning-based approaches have gained significant attention, with several contributions showing competitive performances especially on autonomous driving datasets [17, 44, 47]. We instead focus on efficient indoor 3D perception with ToF. In order to attain overall low power budget for both ToF sensing and depth completion, neural networks must be designed and trained accordingly for efficient inference. In this paper we introduce a carefully designed encoder-decoder CNN for depth completion. We achieve a low memory footprint for weights and activations by means of mixed-precision quantization-aware training and show that the resulting quantized models achieve conspicuous memory size savings for weights and activations, while remaining competitive w.r.t. quality metrics. Our contributions are as follows:

- We propose an efficient depth completion model suitable for low-power sparse ToF sensing. We evaluate our model and other state-of-the-art models on two datasets: the public-domain dataset NYU-Depth v2 [43], and SDS-ST1k [6], a contributed dataset providing accurate sparse ToF and RGB camera simulations.
- We analyze the impact of input pre-processing providing fast initialization; a normals estimation block enabling supervision with ground truth normals; different loss functions and network configurations. Each of these elements contributes to achieving optimal depth map quality for the resulting model.
- We evaluate quantization-aware training strategies to quantize weights and activations of the previous model. To the best of our knowledge, mixed-precision quantization [46] has been used successfully to classification tasks. In this paper we present the first ap-

*Corresponding author: valerio.cambareri@sony.com. XJ, CU, and AS thank Sony Europe B.V. for supporting their internships.

Original p. 2 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).

Original p. 3 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).

Original p. 4 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).

n_f	n_s	MParams	RMSE (mm)	MNS
64	5	50.3	98.7	0.783
32	5	12.6	105.0	0.775
16	5	3.2	181.1	0.751
64	4	12.6	219.2	0.779
64	3	3.1	248.4	0.761
32	4	3.1	274.8	0.767

Table 1. Network tuning on SDS-ST1k (resolution 640×480). We confirm a steady decrease of quality metrics as the feature maps n_f relative to the highest scale and the scales n_s are reduced.

sufficient to retrieve scale. Thus, we minimize

$$\mathcal{L}(\mathcal{D}) := \mathcal{L}_p(\mathcal{D}) + \lambda_n \mathcal{L}_n(\mathcal{D}) \quad (4)$$

which balances between scale-dependent and scale-independent terms using $\lambda_n \in [0, 1]$. This weight depends on the dataset and must be tuned empirically. To do so, we run preliminary tests of our approach (Fig. 2a) on SDS-ST1k under different⁶ $\mathcal{L} := \{\mathcal{L}_{\ell_2}, \mathcal{L}_{\ell_1}, \mathcal{L}_{\ell_1} + \lambda_n \mathcal{L}_n\}$ and $\lambda_n := 10^{-q}$, $q = 0, \dots, 4$, in the same settings described in Sec. 6.1. Indeed, defining the conventional RMSE and Mean Normals Similarity

$$\text{MNS} := \frac{1}{J n_h n_v} \sum_{j=1}^J \sum_{i=1}^{n_h n_v} \langle (N_{\text{GT},j})_i, (\hat{N}_j)_i \rangle \quad (5)$$

as quality metrics we see that large values of λ_n degrade the RMSE, and *vice versa* for MNS. We find the best trade-off at $\lambda_n^{\text{opt}} = 10^{-3}$. As confirmation of this, in Fig. 3b we report one sample error map $\hat{D} - D_{\text{GT}}$ under different $\mathcal{L}$. As we move from $\mathcal{L}_{\ell_2}$ to $\mathcal{L}_{\ell_1}$ the observed error map pattern artefact disappears. Remaining errors at depth map discontinuities are reduced by using the normals loss $\mathcal{L}_n$.

4.5. Network Tuning

We investigated simple strategies to reduce the size of the initial model, such as tuning the $\text{UNet}_{\theta, n_f, n_s}$ module by its number of feature maps n_f relative to the highest scale, and the number of scales n_s . Our preliminary tests, in the same setting as Sec. 6.1 on SDS-ST1k, actually confirm (Tab. 1) that any pair smaller than $(n_f, n_s) = (64, 5)$ leads to a rapid degradation of RMSE and MNS, with the reduction of n_s (shallower network) being generally worse than that of n_f . Thus, we choose to achieve memory size reduction by quantization of the full model with $(n_f, n_s) = (64, 5)$ which, as we will show, allows for a more compact model with limited impact on quality metrics.

5. Quantization-Aware Training

Let $Q(x; \phi)$ be an arbitrary scalar quantizer with parameters ϕ , which maps $x \in \mathbb{R}$ to discrete values $\{q_1, \dots, q_I\}$ represented by b bits, $b : I \leq 2^b$. This quantizer may

⁶We verified in preliminary tests that $\mathcal{L}_{\ell_2} + \lambda_n \mathcal{L}_n$ is systematically worse than the ℓ_1 case and excluded it accordingly.

be applied to both weights and activations to reduce their memory size requirements. This can be done by either post-training quantization or by Quantization-Aware Training (QAT), which generally achieves better accuracy than the former. QAT consists in training the quantized version of a DNN (in this case, CNN) while applying $Q(x; \phi)$ to its weights and/or activations. However, this is challenging as the gradient through the quantizer is zero almost everywhere; a solution to this is the Straight-Through Estimator (STE) [4]. A *symmetric uniform* quantizer $Q_U(x; \phi)$ then maps $x \in \mathbb{R}$ to uniformly quantized values by

$$q = Q_U(x; \phi) := \text{sign}(x) \begin{cases} d \left\lfloor \frac{|x|}{d} + \frac{1}{2} \right\rfloor & |x| \leq q_{\max} \\ q_{\max} & |x| > q_{\max} \end{cases}, \quad (6)$$

where the parameter vector $\phi := [d, q_{\max}, b]^T$ with step size $d \in \mathbb{R}^+$, maximum value $q_{\max} \in \mathbb{R}^+$, and number of bits or *bit width* $b \in \mathbb{N} : b \geq 2$ of the quantized value q .

5.1. Uniform Precision Models

We start by considering *uniform precision* (UP) QAT, *i.e.*, every weight (or activation) of every layer has the same predefined bit width b and QAT optimizes over the remaining parameters $d, q_{\max}$ of ϕ . We conducted extensive experiments to preserve depth completion accuracy while quantizing with (6) the weights and activations in our model. The most effective UP QAT training procedure was obtained by initializing from the pretrained float32 model and using learnable $q_{\max}$ as in the Trained Quantization Threshold (TQT) approach [23]; to stabilize the QAT, we used cosine learning rate decay scheduling [32], and we first trained with RMSprop until convergence (32 epochs in our case), followed by Adam [26] for 20 more epochs.

5.2. Mixed Precision Models

Recent results [36, 46] show that, for a given memory size budget, better accuracy than UP QAT can be achieved using *mixed precision* (MP) QAT, *i.e.*, each layer uses its own bit width *learned at training* to fit a target total network size. Following [46] we parametrize $Q_U(x; \phi)$ with their range and step size d , from which the bit width $b := \lceil 1 + \log_2(q_{\max}/d + 1) \rceil$ is then inferred. With this parametrization and STE [4] gradient, we can use standard stochastic gradient descent methods to learn per-layer optimal bit widths jointly with network parameters. To achieve target network sizes we add to (4) *size constraints* on the weights and activations as penalty terms [5], minimizing

$$\mathcal{L}_{\text{MP}} = \mathcal{L}(\mathcal{D}) + \lambda_W \max(0, S_W)^2 + \lambda_A \max(0, S_A)^2 \quad (7)$$

$$S_x = (\sum_{l=1}^L S_x^l) - S_x^o, S_x^l = N_{x,l} b_{x,l}, S_x^o = \bar{b}_x \sum_{l=1}^L N_{x,l} \quad (8)$$

Original p. 6 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).

Original p. 7 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).

Original p. 8 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).

References

- [1] Gianluca Agresti, Henrik Schafer, Piergiorgio Sartor, Yalcin Incesu, and Pietro Zanuttigh. Unsupervised Domain Adaptation of Deep Networks for ToF Depth Refinement. *IEEE Trans. Pattern Anal. Mach. Intell.*, 2021.
- [2] Yu Bai, Yu-Xiang Wang, and Edo Liberty. ProxQuant: Quantized Neural Networks via Proximal Operators. In *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
- [3] Jonathan T. Barron and Ben Poole. The Fast Bilateral Solver. In *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pages 617–632. Springer International Publishing, 2016.
- [4] Yoshua Bengio, Nicholas Léonard, and Aaron Courville. Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*, 2013.
- [5] Dimitri P. Bertsekas. *Constrained optimization and Lagrange multiplier methods*. Academic Press, 2014.
- [6] Valerio Cambareri. SDS-ST1K Dataset. Technical report, Sony DepthSensing Solutions NV, May 2021.
- [7] Fabien Cardinaux, Stefan Uhlich, Kazuki Yoshiyama, Javier Alonso García, Lukas Mauch, Stephen Tiedemann, Thomas Kemp, and Akira Nakamura. Iteratively Training Look-Up Tables for Network Quantization. *IEEE J. Sel. Top. Signal Process.*, 14(4):860–870, 2020.
- [8] Zhao Chen, Vijay Badrinarayanan, Gilad Drozdov, and Andrew Rabinovich. Estimating depth from rgb and sparse sensing. In *Proc. Eur. Conf. Comput. Vis. (ECCV)*, pages 167–182, 2018.
- [9] Xinjing Cheng, Peng Wang, and Ruigang Yang. Learning depth with convolutional spatial propagation network. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(10):2361–2379, 2019.
- [10] Jungwook Choi, Zhuo Wang, Swagath Venkataramani, Pierce I.-Jen Chuang, Vijayalakshmi Srinivasan, and Kailash Gopalakrishnan. PACT: Parameterized clipping activation for quantized neural networks. *arXiv preprint arXiv:1805.06085*, 2018.
- [11] James Diebel and Sebastian Thrun. An Application of Markov Random Fields to Range Sensing. In Y. Weiss, B. Schölkopf, and J. Platt, editors, *Proc. Advances Neural Inf. Process. Syst. (NeurIPS)*, volume 18. MIT Press, 2005.
- [12] Abdelrahman Eldesokey, Michael Felsberg, Karl Holmquist, and Michael Persson. Uncertainty-Aware CNNs for Depth Completion: Uncertainty from Beginning to End. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020.
- [13] A. Eldesokey, M. Felsberg, and F. Khan. Confidence Propagation through CNNs for Guided Sparse Depth Regression. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(10):2423–2436, Oct. 2020. Place: Los Alamitos, CA, USA Publisher: IEEE Computer Society.
- [14] Epic Games. Unreal Engine (v4.25).
- [15] Pedro F. Felzenszwalb and Daniel P. Huttenlocher. Distance Transforms of Sampled Functions. *Theory of Computing*, 8(19):415–428, 2012.
- [16] Vitor Guizilini, Rares Ambrus, Wolfram Burgard, and Adrien Gaidon. Sparse auxiliary networks for unified monocular depth prediction and completion. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pages 11078–11088, 2021.
- [17] Vitor Guizilini, Rares Ambrus, Sudeep Pillai, Allan Rantos, and Adrien Gaidon. 3D Packing for Self-Supervised Monocular Depth Estimation. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020.
- [18] Song Han, Huizi Mao, and William J. Dally. Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding. In *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016.
- [19] Akio Hayakawa, Masato Ishii, Yoshiyuki Kobayashi, Akira Nakamura, Takuya Narihira, Yukio Obuchi, Andrew Shin, Takuya Yashima, and Kazuki Yoshiyama. Neural Network Libraries: A Deep Learning Framework Designed from Engineers’ Perspectives, 2021.
- [20] Mu Hu, Shuling Wang, Bin Li, Shiyu Ning, Li Fan, and Xiaojin Gong. PENet: Towards Precise and Efficient Image Guided Depth Completion. In *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, pages 13656–13662, 2021.
- [21] Gao Huang, Zhuang Liu, Laurens van der Maaten, and Kilian Q. Weinberger. Densely Connected Convolutional Networks. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2017.
- [22] Saif Imran, Yunfei Long, Xiaoming Liu, and Daniel Morris. Depth Coefficients for Depth Completion. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019.
- [23] Sambhav Jain, Albert Gural, Michael Wu, and Chris Dick. Trained Quantization Thresholds for Accurate and Efficient Fixed-Point Inference of Deep Neural Networks. In I. Dhillon, D. Papailiopoulos, and V. Sze, editors, *Proceedings of Machine Learning and Systems*, volume 2, pages 112–128, 2020.
- [24] M. Jaritz, R. Charette, E. Wirbel, X. Perrotton, and F. Nashashibi. Sparse and Dense Data with CNNs: Depth Completion and Semantic Segmentation. In *Proc. IEEE Int. Conf. 3D Vis. (3DV)*, pages 52–60, 2018.
- [25] Alex Kendall, Matthew Grimes, and Roberto Cipolla. PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization. In *Proc. Int. Conf. Comput. Vis. (ICCV)*, pages 2938–2946, 2015.
- [26] Diederick P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015.
- [27] Jason Ku, Ali Harakeh, and Steven L. Waslander. In Defense of Classical Image Processing: Fast Depth Completion on the CPU. In *2018 15th Conference on Computer and Robot Vision (CRV)*, pages 16–22. IEEE, 2018.
- [28] A. Li, Z. Yuan, Y. Ling, W. Chi, S. Zhang, and C. Zhang. A Multi-Scale Guided Cascade Hourglass Network for Depth Completion. In *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, pages 32–40, 2020.
- [29] Fengfu Li, Bo Zhang, and Bin Liu. Ternary weight networks. *arXiv preprint arXiv:1605.04711*, 2016.
- [30] Zhi-Gang Liu and Matthew Mattina. Learning Low-precision Neural Networks without Straight-Through Estimator (STE). In *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, pages 3066–3072, 2019.

Original p. 10 is not reproduced in this excerpt.
Full article: IEEE Xplore, <https://doi.org/10.1109/CVPRW56347.2022.00302> (© IEEE).